

Epic 4: Data Preprocessing and Label Encoding

Selecting Dependent and Independent Variables

Data preprocessing is an important step in machine learning.

The dataset contains multiple columns with different information.

Independent variables are the input features used for prediction.

The dependent variable is the target output of the model.

Feature selection improves model performance.

It reduces unnecessary columns and noise.

Relevant variables help algorithms learn better.

For the HDI project, life expectancy, education, and income are common inputs.

The HDI value is the dependent variable.

Proper variable selection avoids overfitting.

It improves training speed.

Data understanding is essential.



Checking and Handling Null Values

Missing values reduce dataset quality.

Null values occur because of incomplete data.

Checking null values is performed before training.

Pandas provides `isnull()` and `sum()` methods.

Rows with excessive missing values may be removed.

Numerical values are filled with mean or median.

Categorical values are filled with mode.

Proper handling prevents errors.

Clean datasets improve prediction accuracy.

Validation is required after filling values.

Consistency is maintained.

The processed dataset becomes reliable.



Label Encoding

Machine learning models require numerical inputs.

Categorical values are converted into numbers.

Label Encoding assigns integer values.

It is useful for ordinal categories.

Scikit-learn provides LabelEncoder.

Encoding improves compatibility.

The transformed dataset is easier to process.

Encoded data supports model training.

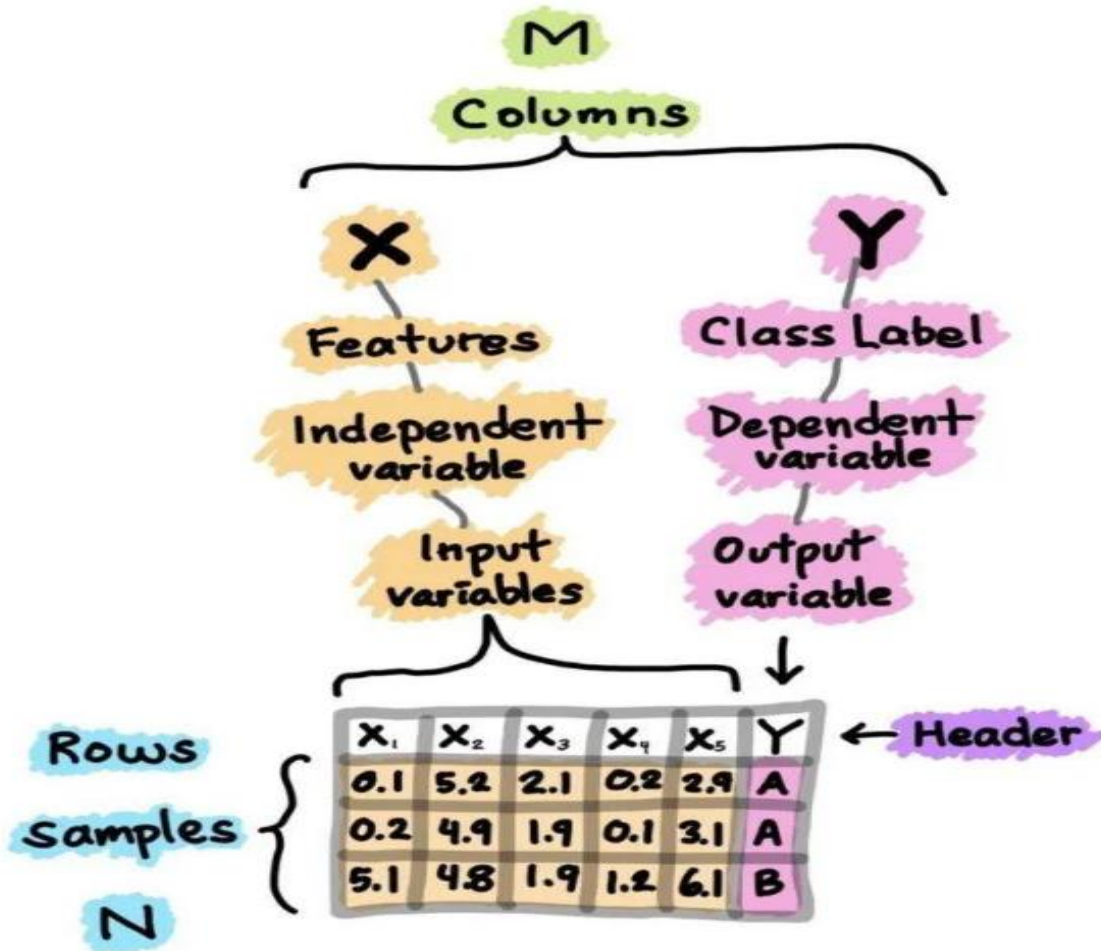
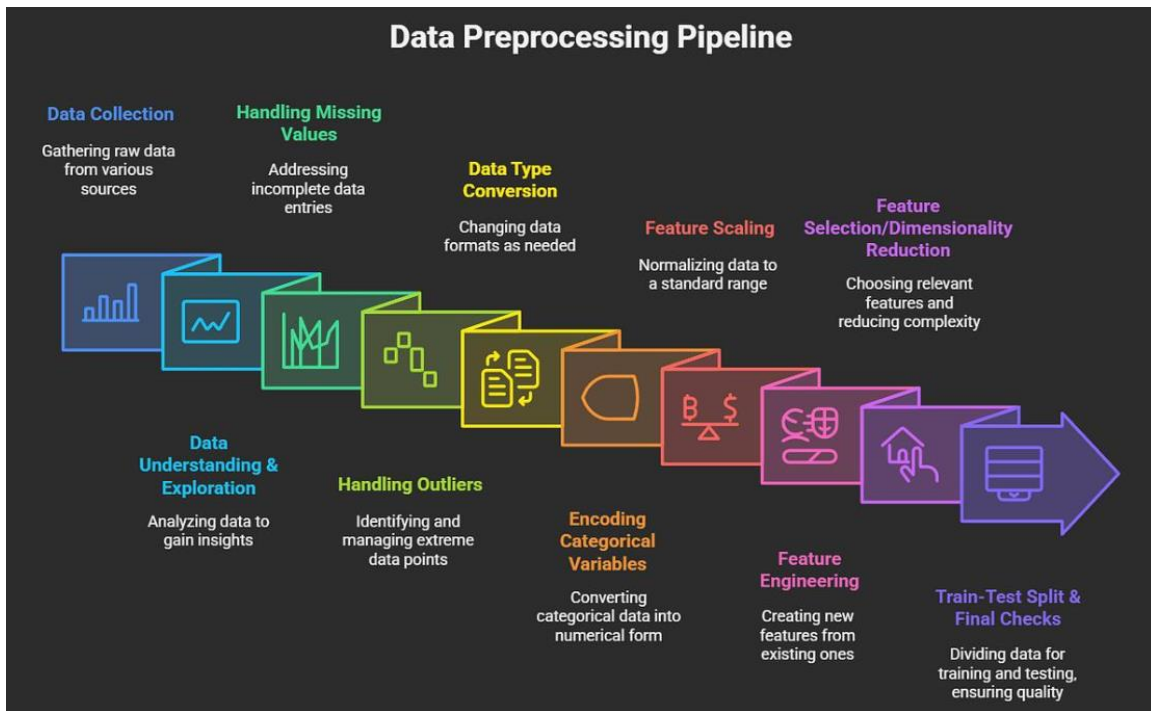
Preprocessing increases efficiency.

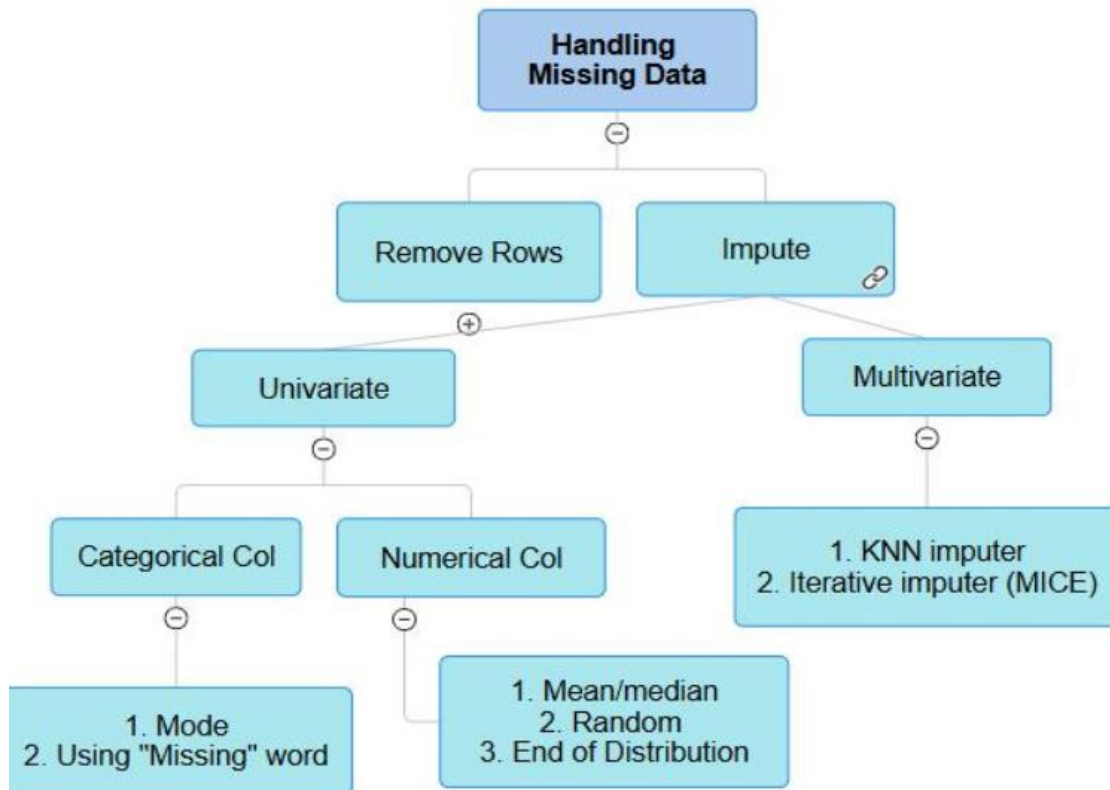
Correct encoding avoids errors.

These steps improve prediction quality.

This completes preprocessing.







original dataset

x ₁	x ₂	y
5	8	calabar
9	3	uyo
8	6	owerri
0	5	uyo
2	3	calabar
0	8	calabar
1	8	owerri

LabelEncoder



```

{
  "calabar" ---> 0
  "owerri" ---> 1
  "uyo" ---> 2
}
  
```

dataset with encoded labels

x ₁	x ₂	y
5	8	0
9	3	2
8	6	1
0	5	2
2	3	0
0	8	0
1	8	1

